



JUL 03 2014

Docket Nos.: 52-025
52-026

ND-14-0748
10 CFR 50.90
10 CFR 52.63

U.S. Nuclear Regulatory Commission
Document Control Desk
Washington, DC 20555-0001

Southern Nuclear Operating Company
Vogtle Electric Generating Plant Units 3 and 4
Request for License Amendment and Exemption Regarding
Containment Internal Structural Module Design Details (LAR-14-001)

Ladies and Gentlemen:

In accordance with 10 CFR 50.90, Southern Nuclear Operating Company (SNC), the licensee for Vogtle Electrical Generating Plant (VEGP) Units 3 and 4, requests an amendment to Combined License (COL) Numbers NPF-91 and NPF-92, for VEGP Units 3 and 4, respectively. Pursuant to the provisions of 10 CFR 52.63(b)(1), an exemption from elements of the design as certified in the 10 CFR Part 52, Appendix D, design certification rule is also requested for the plant-specific DCD Tier 1 material departures.

The requested amendment proposes to depart from Tier 2* information in the Updated Final Safety Analysis Report (UFSAR), plant-specific Tier 1 and corresponding COL Appendix C information, and involved UFSAR Tier 2 information to address changes in the UFSAR and design documents related to containment internal structural wall module design details.

The description, technical evaluation, regulatory evaluation (including the significant hazards consideration determination), and environmental considerations for the proposed changes in the License Amendment Request (LAR) are contained in Enclosure 1 to this letter. Enclosure 2 provides the background and supporting basis for the requested exemption. Enclosure 3 provides markups depicting the requested changes to the licensing basis documents. Enclosure 4 includes markups depicting the requested changes to the UFSAR figures which are withheld from public disclosure as Security-Related Information, in accordance with 10 CFR 2.390(d).

This letter contains no regulatory commitments.

SNC requests NRC staff approval of the license amendment and associated exemption by November 14, 2014, to support placement of concrete in internal structural module CA05. Approval by this date will allow sufficient time to implement the licensing basis changes. SNC expects to implement this proposed amendment (through incorporation into the licensing basis documents, e.g., the Updated Final Safety Analysis Report) within 30 days of approval.

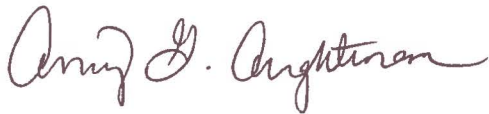
In accordance with 10 CFR 50.91, SNC is notifying the State of Georgia of this LAR by transmitting a copy of this letter and enclosures to the designated State Official.

Should you have any questions, please contact Mr. D. Rick Graham at (205) 992-5808.

Ms. Amy G. Aughtman states that: she is a Licensing Manager of Southern Nuclear Operating Company; she is authorized to execute this oath on behalf of Southern Nuclear Operating Company; and to the best of her knowledge and belief, the facts set forth in this letter are true.

Respectfully submitted,

SOUTHERN NUCLEAR OPERATING COMPANY



A. G. Aughtman

AGA/DRG/kms

Sworn to and subscribed before me this 3rd day of July, 2014

Notary Public: Kristin Marie Seibert

My commission expires: August 16, 2016



- Enclosures:
- 1) Vogtle Electric Generating Plant (VEGP) Units 3 and 4 – Request for License Amendment Regarding Containment Internal Structural Wall Module Design Details (LAR-14-001)
 - 2) Vogtle Electric Generating Plant (VEGP) Units 3 and 4 – Request for Exemption Regarding Containment Internal Structural Wall Module Design Details (LAR-14-001)
 - 3) Vogtle Electric Generating Plant (VEGP) Units 3 and 4 – Proposed Changes to Licensing Basis Documents (LAR-14-001)
 - 4) Vogtle Electric Generating Plant (VEGP) Units 3 and 4 – Proposed Changes to Licensing Basis Documents (LAR-14-001) - **Withheld Information**

cc:

Southern Nuclear Operating Company / Georgia Power Company

Mr. S. E. Kuczynski (w/o enclosures)

Mr. J. A. Miller

Mr. D. A. Bost (w/o enclosures)

Mr. B. L. Ivey

Mr. M. D. Rauckhorst (w/o enclosures)

Mr. J. T. Gasser (w/o enclosures)

Mr. D. H. Jones (w/o enclosures)

Mr. J. R. Johnson (w/o enclosures)

Mr. T. E. Tynan (w/o enclosure 4)

Mr. D. M. Lloyd

Mr. B.H. Whitley

Mr. C. R. Pierce

Mr. D. L. Fulton

Mr. M. J. Yox

Mr. J. C. Harrelson

Ms. A. G. Aughtman

Mr. W. A. Sparkman

Mr. J. P. Redd

Document Services RTYPE: VND.LI.L00

File AR.01.02.06

Nuclear Regulatory Commission

Mr. V. M. McCree (w/o enclosures)

Mr. M. Delligatti (w/o enclosures)

Mr. L. Burkhardt (w/o enclosures)

Mr. D. H. Jaffe

Mr. R. G. Joshi

Ms. D. L. McGovern

Mr. B. M. Bovol

Ms. M. A. Sutton

Mr. M. E. Ernestes

Mr. G. Khouri

Mr. L. M. Cain

Mr. J. D. Fuller

Mr. C. B. Abbott

Mr. C. Huffman

State of Georgia

Mr. J. H. Turner (w/o enclosure 4)

Oglethorpe Power Corporation

Mr. M. W. Price (w/o enclosure 4)

Mr. K. T. Haynes (w/o enclosure 4)

Municipal Electric Authority of Georgia

Mr. J. E. Fuller (w/o enclosure 4)

Mr. S. M. Jackson (w/o enclosure 4)

Dalton Utilities

Mr. D. Cope (w/o enclosure 4)

CB&I

Mr. J. Simmons (w/o enclosures)

Ms. K. Stoner (w/o enclosures)

Mr. C. A. Castell

Westinghouse Electric Company, LLC

Mr. T. C. Geer (w/o enclosures)

Mr. S. W. Gray (w/o enclosures)

Mr. F. G. Gill

Mr. P. A. Russ

Mr. G. F. Couture

Mr. M. Y. Shaqqo

Other

Mr. R. W. Prunty, Bechtel Power Corporation

Ms. K. K. Patterson, Tetra Tech NUS, Inc. (w/o enclosure 4)

Dr. W. R. Jacobs, Jr., Ph.D., GDS Associates, Inc. (w/o enclosure 4)

Mr. S. Roetger, Georgia Public Service Commission (w/o enclosure 4)

Ms. S. W. Kernizan, Georgia Public Service Commission (w/o enclosure 4)

Mr. K. C. Greene, Troutman Sanders (w/o enclosure 4)

Mr. S. Blanton, Balch Bingham

Ms. A. Rice, South Carolina Electric & Gas Company

Mr. D. Kersey, South Carolina Electric & Gas Company

Mr. B. Kitchen, Duke Energy

Mr. S. Franzone, Florida Power & Light

Southern Nuclear Operating Company
Vogtle Electric Generating Plant (VEGP) Units 3 and 4

ND-14-0748

Enclosure 1

Request for License Amendment Regarding
Containment Internal Structural Wall Module Design Details
(LAR-14-001)

Table of Contents

1. Summary Description
2. Detailed Description
3. Technical Evaluation
4. Regulatory Evaluation
 - 4.1 Applicable Regulatory Requirements/Criteria
 - 4.2 Precedent
 - 4.3 Significant Hazards Consideration Determination
 - 4.4 Conclusions
5. Environmental Considerations

1. Summary Description

The purpose of this License Amendment Request (LAR) is to address proposed changes related to the design details of the in-containment structural wall modules (CA01, CA02, and CA05). These design changes are a result of design finalization of the modules, fabrication experience in the shop environment, and the comparison of text and figures presented in the Updated Final Safety Analysis Report (UFSAR). These changes include the following: 1) providing additional detail to describe the mechanical connections between the internal containment structural module steel faceplates and the base concrete, 2) allowing for increases in the thickness of the structural wall module faceplates, 3) identifying changes to the wall thicknesses for portions of some internal containment structural wall modules, and 4) identifying the use of steel plates, structural shapes, reinforcement bars, or tie bars between the faceplates of the structural wall modules, where needed to meet applicable code requirements.

The proposed changes to UFSAR Tier 2 information and the involved plant-specific Tier 1 and associated COL Appendix C information allow for the use of thicker than nominal faceplates to accommodate local demand or connection loads in certain areas without the use of overlay plates or additional backup structures. Additional proposed changes to Tier 2 and involved Tier 2* information allow: 1) a means of connecting the structural wall modules to the base concrete via use of structural shapes, reinforcement bars, and shear studs extending horizontally from the structural module faceplates and embedded during concrete placement as an alternative to the use of embedment plates and vertically oriented reinforcement bars, 2) a variance in structural module wall thicknesses from the thicknesses identified in UFSAR Figure 3.8.3-8, "Structural Modules – Typical Design Details," for some walls that separate equipment spaces from personnel access areas, and 3) the use of steel plates, structural shapes, reinforcement bars, or tie bars between the module faceplates, as needed to support localized loads and ensure compliance with applicable codes.

2. Detailed Description

The proposed revisions to the UFSAR are needed to reflect proposed design changes related to the internal containment structural wall module faceplate nominal thickness and wall thicknesses, to the mechanical connection between the in-containment structural module steel faceplate and the base concrete, and to the use of steel plates, structural shapes, reinforcement bars, or tie bars between the faceplates to address out-of-plane loads. These proposed changes are described below. The Tier 2 text changes identified in this document include involved Tier 2* changes in Subsections 3.8.3.1.3, "Structural Wall Modules," 3.8.3.5.8.1, "Structural Wall Modules," and 3.8.3.8, "Construction Inspection," in Table 3.8.3-3, "Definition of Critical Locations and Thicknesses for Containment Internal Structures," and Table 3.8.3-5, "Design Summary of South Wall of Steam Generator

Compartment Design Loads, Load Combinations, and Comparison to Acceptance Criteria Mid-Span at Mid-Height,” and in Figure 3.8.3-1, “Structural Modules in Containment Internal Structures,” Figure 3.8.3-8, “Structural Modules – Typical Design Details,” and Figure 3.8.3-18, “Location of Structural Wall Modules.”

The applicability of these changes to the affected modules is as shown in the table below.

Location	In-Containment Structural Wall Module		
	CA01	CA02	CA05
Structural Module Connection to the Base Concrete	X		X
Structural Module Faceplate Thickness	X		X
Structural Module Wall Thickness	X		X
Structural Module Out-of-Plane Reinforcement	X	X	X

Structural Module Connection to the Base Concrete

UFSAR Figure 3.8.3-8 (Sheet 2) identifies a typical mechanical attachment of the structural module faceplate to the base concrete, showing the module attachment to an embedment plate with vertically oriented reinforcement bars that are embedded in the concrete. UFSAR Figure 3.8.3-8 (Sheet 2) illustrates a module wall to base concrete connection where base concrete placement occurs after the module is set. In these cases, a single module faceplate extends to a greater depth into the base concrete than the opposite faceplate due to plant layout constraints. The proposed changes to the text in UFSAR Subsections 3.8.3.1.3 and 3.8.3.5.3 add additional details for these connections. Specifically, the additional details identify the use of horizontally oriented reinforcement bars, structural shapes, and shear studs mechanically attached or welded to the lower module faceplate, and embedded in the base concrete to transfer forces horizontally from the wall module faceplate to the base concrete. These connections are designed according to the applicable sections of ACI 349 and AISC N690. In addition, the proposed modifications permit that shear-friction construction joint reinforcement, which is not attached to the module faceplate, may be included in construction joints at the transition between the base concrete and concrete in the modules. The shear-friction construction joint reinforcement is not

connected to the module faceplates and is not credited with connecting the module to the base concrete.

Each of these proposed alternatives are in compliance with ACI 349 requirements, as identified in UFSAR Subsection 3.8.3.1.3. These changes are applicable to in-containment structural modules CA01 and CA05.

Structural Module Faceplate Thickness

Supports for piping, electrical raceways, equipment, and components are attached to the module structures by welding the supports directly to the structural module faceplate, or to overlay plates embedded inside the module walls. The loads from the welded supports are transferred to the structural module faceplates and shear studs or backup structure elements into the concrete. Overlay plates, where used, transmit loads from attached commodities directly to the concrete inside the module wall. The UFSAR identifies the nominal thickness of the structural module faceplates as 0.5 inch in text, figures, and tables. In some locations in the CA01 and CA05 structural modules, faceplates with thicknesses up to 1.5 inches are used to support local demand or connection loads without the use of overlay plates. In several areas within the upper portions of the CA01 steam generator compartments, 3.0 inch faceplates are used in local areas to support steam generator lateral support loads. In some locations, overlay plates are used, which also add thickness at that location. The additional thickness is applicable to the outer surface of the module and, as a result, the overall module wall thickness is correspondingly greater in these areas. Faceplates with thicknesses greater than nominal are used on the CA01 structural module (UFSAR Figure 3.8.3-14, Sheet 1), as well as CA05. Overlay plates are used in the CA01, CA02, and CA05 structural modules. Thus, the UFSAR is revised in several places, as detailed below, to identify these variations in structural module faceplate thickness.

Structural Module Wall Thickness

UFSAR Figure 3.8.3-8 (Sheet 1) specifies that the wall thickness for the structural wall modules is either a nominal 2'-6" or a nominal 4'-0". This thickness requirement was intended for the major structural walls of the structural modules such as the primary and secondary shield walls around the reactor vessel and steam generator, and the walls of the refueling canal. However, there are other module walls, such as walls that separate equipment spaces from personnel access areas, which can satisfy their associated structural design and shielding requirements, using wall thicknesses different than the values currently specified in UFSAR Figure 3.8.3-8 (Sheet 1). In addition, some walls require thicknesses greater than 4'-0" in order to satisfy shielding requirements. The maximum thickness used in the structural wall modules is 4'-6" as shown in Figure 3.8.3-1 (Sheet 2), and the minimum thickness used in the structural wall modules is 1'-6" as shown in Figure 3.8.3-1 (Sheet 2).

Thus, it is proposed to revise UFSAR Figure 3.8.3-8 (Sheets 1 and 2) to:

- Add a note stating that some module walls that separate equipment spaces from personnel access areas have wall thicknesses that differ from the dimensions shown,
- Provide a reference to UFSAR Figure 3.8.3-1 (Sheets 2 and 4), which currently identify, and are being revised to identify more of the structural module CA01 and CA05 wall thicknesses; and make conforming changes to the text in UFSAR Section 3.8.3.1.3,
- Identify that the affected module wall thicknesses are determined based on radiation protection considerations, and
- Identify that the wall modules are designed to the applicable ACI 349 and AISC N690 code requirements.

The proposed revisions include changes to UFSAR Figure 3.8.3-1 (Sheets 1 - 5), and Figure 3.8.3-18, to add notes to define a specific elevation for the figures, and revise the module outlines as necessary to accurately reflect the module outlines at the newly specified elevations.

The proposed revisions include changes to UFSAR Figure 3.8.3-1 (Sheet 2) to add wall thicknesses for wall sections that vary from the nominal wall thicknesses, and to show a notch inside the northeast corner of structural module CA05 to denote a region of the module where the thickness was reduced to accommodate equipment in that area, which transitions at higher elevations to full CA05 wall thickness.

The proposed revisions include changes to UFSAR Figure 3.8.3-1 (Sheet 4) and UFSAR Figure 3.8.3-18 to include the addition of a shielding wall on the northeast side of CA01 to be consistent with the newly specified elevation of 107'-2", and consistent with the UFSAR Figure 1.2-7, "Nuclear Island General Arrangement Plan at Elevation 107'-2" & 111'-0".

The proposed revisions include changes to UFSAR Figure 3.8.3-1 (Sheet 4) to add wall thicknesses for wall sections that vary from the nominal wall thicknesses.

These changes are applicable to in-containment structural modules CA01 and CA05.

Structural Module Out-of-Plane Reinforcement

Certain portions of the structural modules experience out-of-plane loads and require additional reinforcement according to the applicable sections of AISC N690 and ACI 349. In order to address these areas of localized out-of-plane loads, it is proposed to add steel plates, structural shapes, reinforcement bars, or tie bars between the faceplates. This additional reinforcement would be attached at both ends to the faceplates and embedded in the concrete, and added to ensure compliance with code requirements. As identified below, the provision to provide additional reinforcement to support demand from out-of-plane loads is being added to the UFSAR in several locations. These changes are applicable to in-containment structural modules CA01, CA02, and CA05.

Licensing Basis Change Descriptions

The affected text, tables, and figures are proposed to be modified as discussed below (grouped by change category) and shown in Enclosures 3 and 4.

Structural Module Connection to the Base Concrete

1. UFSAR Subsection 3.8.3.1.3 is revised to add that shear-friction construction joint reinforcement, which is not attached to the module faceplate, may be included in construction joints at the transition between the base concrete and concrete in the modules in accordance with ACI 349 requirements.
2. UFSAR Subsections 3.8.3.1.3 and 3.8.3.5.3 are revised to collectively add that i) shear studs, structural shapes, and horizontally oriented reinforcement bars mechanically connected or welded to the module faceplates may be used to anchor the modules to the base concrete, ii) where shear lugs are used, they are designed in accordance with ACI 349, iii) where horizontal bars are attached perpendicular to the faceplate, the bars act as shear studs to develop the faceplate, and iv) where reinforcement bars are used to further detail base concrete at module walls with offset elevations, or are attached to the edge of a faceplate at an elevation offset, the development satisfies the requirements of ACI 349. This information is designated Tier 2*.

Structural Module Faceplate Thickness

1. UFSAR Subsections 3.8.3.1.3 and 3.8.3.8, Table 3.8.3-3, Table 3.8.3-4, Table 3.8.3-5, and Figure 3.8.3-8 (Sheets 1 and 2) are revised to collectively identify that in some locations, the faceplate thickness of the structural modules is greater than the 0.5 inch nominal faceplate thickness up to a thickness of 1.5 inches to support local demand associated with equipment support loads, and to support steam generator lateral loads, the faceplate thickness is increased to 3.0 inches. These additions have been designated Tier 2*. The beginning of the second paragraph of Section 3.8.3.1.3 is also re-arranged in this area to better organize the information and provide continuity to the structural module faceplate, and truss and diaphragm discussions.
2. UFSAR Table 3.8.3-3 is revised to correct a typographical error by removing the leading period for the 0.01 values in all three rows for the column titled "Thickness of Surface Plates Provided." An additional change is made to that table to remove the word "maximum" from the title of the fifth column.
3. UFSAR Subsection 3.8.3.5.3 is revised to add a statement that variations in the module design details and design elements used are described in the Figure 3.8.3-8 notes and in subsections 3.8.3.1 and 3.8.3.1.3.
4. Plant-specific Tier 1 and associated COL Appendix C Table 3.3-1, Note 3 is revised to add that where structural wall module faceplates thicker than the nominal 0.5 inches or overlay plates are used in the construction of the structural wall modules, the wall thicknesses in the area of the thicker faceplates and overlay plates are larger than indicated in the table, by the amount of faceplate and/or overlay plate thickness increase over the nominal 0.5 inches.

Structural Module Wall Thickness

1. UFSAR Tier 2* Subsection 3.8.3.1.3 is revised, and a note added to Figure 3.8.3-8 (Sheets 1 and 2) to state that some module walls that separate equipment spaces from personnel access areas have wall thicknesses different than the dimensions shown on Figure 3.8.3-8, and that Figure 3.8.3-1 (Sheets 2 and 4) identify the CA01 and CA05 module wall thickness differences. Subsection 3.8.3.1.3 is also revised to identify

that module wall thickness increases due to increases in faceplate thickness is below the level of detail added to UFSAR Figure 3.8.3-1.

2. UFSAR Tier 2* Figure 3.8.3-1 (Sheets 1 - 5), and Tier 2* Figure 3.8.3-18, are revised to add notes to identify specific elevations for each sheet, and revise the drawings as necessary to reflect the wall module configurations at those elevations. These changes include:

- Revision of Figure 3.8.3-1 (Sheet 2) as follows:

CA01

- Removal of the south wall of the refueling cavity, and darkened the remaining line of the trapezoidal corner assemblies
- Identification of the east wall of the east steam generator compartment as a single faceplate wall,

CA05

- Addition of an extension on the north end of the north wall of the CA05 module to identify an extension consisting of a single faceplate wall, and
 - Insertion of a “notch” on the south side of the northeast corner of the module due to a transition in the CA05 module in that location,
 - Addition of wall thickness dimensions to the CA05 module, and
 - Identification of the west end of the CA05 module as a single faceplate wall.
- Revision of Figure 3.8.3-1 (Sheet 3) to identify an extension of the CA05 module north wall further north, as a single faceplate wall,
 - Revision of Figure 3.8.3-1 (Sheet 4) and Figure 3.8.3-18 to identify the addition of a shielding wall on the northeast side of CA01 near the east steam generator compartment opening to be consistent with Figure 1.2-7, “Nuclear Island General Arrangement Plan at Elevation 107'-2" & 111'-0",” and

- Revision of Figure 3.8.3-1 (Sheet 4) to add wall thickness dimensions for the CA01 module walls.

Structural Module Out-of-Plane Reinforcement

1. UFSAR Subsection 3.8.3.1.3 is revised to add that steel plates, structural shapes, reinforcement bars, or tie bars are installed between the structural wall module faceplates and embedded in the concrete as additional structural elements that support out-of-plane loads and structural integrity. This reinforcement is designed to the applicable requirements of ACI 349 and AISC N690.
2. UFSAR Subsection 3.8.3.5.3.5 is revised to add that additional structural capacity is provided by steel plates, structural shapes, reinforcement bars, and tie bars installed between the faceplates as described in UFSAR subsection 3.8.3.1.3, and that these structural members are designed according to the applicable requirements of AISC N690 and ACI 349.
3. UFSAR Tier 2* Subsection 3.8.3.5.8.1 is revised to add that steel plates, structural shapes, reinforcement bars, or tie bars are used in other locations as needed between the structural module faceplates to address out-of-plane loads as described in subsection 3.8.3.1.3.

3. Technical Evaluation

Structure, System, Component and/or Analysis Descriptions

The primary function of the containment internal structures is to provide support, protection, and separation for the seismic Category I mechanical and electrical equipment located inside those structures. These structures are designed to meet seismic Category I requirements as defined in Regulatory Guide 1.29.

The containment internal structures are those concrete and steel structures inside (not part of) the containment pressure boundary that support the reactor coolant system components and related piping systems and equipment. The walls of these modules provide support for floors and other steel structures, component and piping supports, and other attachments. Structural wall modules are used for the primary shield wall around the reactor vessel, for the wall between the vertical access and the Chemical and Volume Control System (CVS) room, for the secondary shield walls around the steam generators and pressurizer, for the

east side of the in-containment refueling water storage tank, and for the refueling cavity. Structural wall modules consist of steel faceplates with shear studs connected by steel trusses, and in localized areas, additional reinforcement. Following installation of the structural module within the containment building, the hollow wall structures are filled with concrete, forming a portion of the structural walls of the containment internal structures.

The containment internal structural modules are seismic Category I structures and are designed for dead, live, thermal, pressure, safe shutdown earthquake, and loads due to postulated pipe breaks.

Supporting Technical Details

Structural Module Connection to the Base Concrete

For the attachment method shown in UFSAR Figure 3.8.3-8 (Sheet 2), mechanical connectors, typically bolts or welds, are used between the module faceplate and the embedment wherever the embedment or anchor is set in the concrete prior to setting the module. These embedments include vertical reinforcing bars which extend into the base concrete. This is currently shown in UFSAR Figure 3.8.3-8 (Sheet 2).

As also shown in UFSAR Figure 3.8.3-8 (Sheet 2), in some cases a single module faceplate extends to a lower elevation than the opposite faceplate due to plant layout constraints. This offset module base configuration occurs in various locations for modules CA01, CA02, and CA05. In the CA01 module, this offset configuration occurs in nearly every submodule, with varying degrees of offset. The CA02 and CA05 modules also have this offset base configuration for most of these module perimeters.

The connection of the lower faceplate in portions of the CA01 and CA05 module to the base concrete may be detailed to transfer forces from the faceplate to the base concrete using horizontally oriented reinforcement bars, structural shapes, shear studs, or shear lugs which are mechanically attached or welded to the lower module faceplate, and are embedded in the base concrete. This connection is designed according to the applicable sections of ACI 349 and AISC N690. Similar to the primary attachment method described above, this proposed alternate method of attachment provides a direct load path between the steel faceplates to the reinforcement bars in the base concrete as required by the note under the figure in UFSAR Figure 3.8.3-8 (Sheet 2). For this alternate attachment method, the structural module is anchored using horizontally oriented deformed bars, headed studs, structural shapes, shear lugs, and other anchors directly connected to the module. These items are embedded in the concrete after the module is set and, as a result, a separate mechanical connection is not required at the base as with the primary attachment method. This is consistent with UFSAR Figure 3.8.3-8 (Sheet 3) which shows the use of studs and deformed bars to anchor a module in the concrete. Where horizontal bars are attached perpendicular to the faceplate, the bars act as shear studs. In other configurations, such as

an offset in faceplate elevation, horizontal bars attached to the faceplate are developed in accordance with ACI 349. In cases where a step in elevation occurs, horizontal bars attached to the edge of the faceplate may be used, in which case, the bars would be fully developed in accordance with ACI 349. Consequently, these anchors transfer load from the module to the concrete regardless of the orientation of the anchor. The transfer of loads from the anchor within the concrete provides a direct connection between the steel module and the base concrete as required by the note in UFSAR Figure 3.8.3-8 (Sheet 2) because the anchors are directly connected to the module faceplates using mechanical or welded connections. In some cases, the structural wall module may need additional detail, requiring the addition of reinforcement bar at the location that is not connected to the module faceplate. In these cases the reinforcement bar is also developed in accordance with ACI 349. The design of this connection uses either shear friction criteria described in ACI 349 Section 11.7 or ACI 349 Appendix B for anchoring to concrete. Wherever reinforcement bars are used, the development requirements of ACI 349 apply.

The horizontally oriented connection reinforcing bar dowels are sized in accordance with the shear-friction provisions of ACI 349 Section 11.7 based on design demand consisting of the applicable combination of the tension forces and in-plane and out-of-plane shear forces. Alternatively, structural shapes such as angles or channels welded to the faceplates may be used to provide shear transfer by consideration of the bearing strength of the concrete in combination with tension anchors in accordance with ACI 349 Appendix B requirements. Compression forces are transferred through the concrete within the composite section directly to the reinforced concrete. Evaluations are performed in accordance with ACI 349 Appendix B to preclude concrete failure modes in tension and shear. Additional reinforcement where required for detailing base concrete at module walls where offsets in faceplate elevation occur is provided in accordance with ACI 349.

This alternate method of attachment satisfies the applicable code requirements of ACI 349 for anchorage in the concrete. The mechanical design and acceptance criteria for the structural module walls as specified in UFSAR Section 3.5 for missile protection, UFSAR Section 3.6 for protection against dynamic effects, UFSAR Section 3.7 for seismic design, and UFSAR Section 3.8 for design of Category I structures are unchanged by the use of this alternate attachment method; consequently, those requirements are met by this alternate attachment method.

It should be noted that, for this alternate attachment method, a lap splice arrangement between the faceplates and reinforcement bar is not used to transfer the load from the module. These items are embedded in the concrete after the module is set, and separate mechanical connection at the base is not required. These items are connected to a single sided portion of the module, extend into the base concrete, and are not located between module walls. The development, embedment requirements, and design of these elements satisfy the applicable portions of ACI 349 and AISC N690.

The change to UFSAR Subsection 3.8.3.1.3 regarding shear-friction construction joint reinforcement is proposed to clarify that the purpose of the shear-friction reinforcement is to satisfy ACI 349 requirements for the concrete construction joint rather than to anchor the module. The information concerning shear-friction reinforcement is added because the reinforcement arrangement used may appear to be a connection between the modules and base concrete using a lap splice type of connection.

Structural Module Faceplate Thickness

As discussed above, the in-containment module design may incorporate faceplates in some locations in structural module CA01 and CA05 that are thicker than the nominal thickness. In addition, overlay plates are attached in some locations to structural modules CA01, CA02, and CA05 to support equipment attachment. Thicker than nominal faceplates are used in localized areas of higher loads. These include areas of increased loading due to equipment supports, such as pipe and component supports or in some connections to floor modules, and permit larger attachment or connection loads without the use of overlay plates or additional backup structures. A thicker faceplate is also used in the area inside the northeast corner of the CA05 structural module where it transitions at lower elevations from 2'-0" (where the wall thickness is reduced to accommodate equipment) to 4'-6".

The sections of thicker faceplates primarily consist of faceplates 1.0 inch and 1.5 inches in thickness. However, in a few locations, in the upper portions of the CA01 structural module steam generator enclosure area, small plates with a thickness of 3 inches are provided to meet local demand for the steam generator lateral supports. For the CA01 structural module, approximately 10 percent of the faceplate area has a faceplate thickness of 1.0 inch, approximately 5 percent of the faceplate area has a faceplate thickness of 1.5 inches, and less than 0.5 percent of the faceplate area has a thickness of 3.0 inches.

The use of thicker faceplates in selected areas provides several design and fabrication benefits. Use of thicker faceplates provides flexibility in selecting the final design location of required attachments, provides fabrication advantages where connections to the faceplates require a large amount of welding and associated heat input, and reduce the use of overlay plates. Use of thicker faceplates in areas of higher loads simplify fabrication by reducing or eliminating drilling, fabrication, and welding operations associated with installing overlay plates and backup structures. The use of thicker faceplates in these areas also provide more effective load transfer to a backup structure or other internal embedments where such an arrangement is included in the design.

Because the thicker faceplates are used only in local regions, they do not have an adverse effect on the structural or seismic analysis of the structural modules. Evaluations were performed to assess the impact of the increase in faceplate thickness in the applicable areas of the affected structural modules. These evaluations determined that the impact on the calculated loads for the critical locations for the in-containment structural modules

documented in UFSAR Table 3.8.3-5 is insignificant, the critical locations as defined in Table 3.8.3-3 have not changed, and the required thickness of the surface plates is not affected. Thus, no changes to the values in these tables are proposed with this LAR. An assessment was also performed to determine the impact of increasing the faceplate thickness on the global seismic model. It has been confirmed that the changes in faceplate thickness for the affected structural modules have an insignificant impact on the global seismic model, and thus no changes to that model are proposed with this LAR. The use of thicker faceplates is in compliance with the applicable code requirements contained in AISC N690 and ACI 349, including the applicable acceptance criteria.

Where overlay plates are used, they function as load attachment and distribution elements. The overlay plates do not affect the structural wall module stiffness used in the global model and do not participate in resisting the internal wall module forces from the global model, including the global seismic analysis.

For the in-containment structural wall modules CA01, CA02, and CA05, duplex steel is used for surfaces in contact with borated water, i.e., wetted surfaces that form portions of the In-Containment Refueling Water Storage Tank (IRWST), the reactor vessel cavity, and the refueling cavity. The duplex faceplates have been determined to have sufficient capacity to meet all structural demand. The CA01 to CA03 connection joints use duplex steel, in the form of T-sections fabricated from 1.5 inch duplex steel. Duplex steel is used in these locations because of the exposure of these connection locations to borated water.

In accordance with AWS D1.1, "Structural Welding Code – Steel," welded transitions between plates of different thicknesses are tapered at a minimum transition slope of 1 in 2.5 to provide a smooth tension stress path through the transition area, and welded using complete joint penetration (CJP) welds. In addition, CJP welds require 100% magnetic particle or liquid penetrant testing, with an additional 10% of the welds subject to volumetric inspection. CJP welds in duplex plate in contact with borated water also require 100% dye penetrant (PT) testing of the final pass. Since the duplex material is not included in the AWS D1.6, "Structural Welding Code – Steel," code as a prequalified material, the fabricator and constructor must develop and qualify the welding procedure that will be used to weld the duplex as required by the AP1000 welding specification for duplex steel plate, and AWS D1.6 Sections 3, 4 and 5.

Table 3.8.3-3 is revised to delete the word "Maximum" from the title of the fifth column "Required Thickness of Surface Plates (inches)". "Maximum" is deleted from the Table 3.8.3-3 fifth column title, as the values presented in that column represent the minimum required thickness of surface plates for operating and design basis loads as identified in Note 3 of Table 3.8.3-3, as well as in subsection 3.8.3.8.

Structural Module Wall Thickness

The proposed change to UFSAR Subsection 3.8.3.1.3 is to clarify that some CA01 and CA05 module walls that separate equipment spaces from personnel access areas have wall thicknesses different than the dimensions shown in Figure 3.8.3-8. For example, the 1'-6" and 1'-6.5" stub walls depicted on the CA01 module in Figure 3.8.3-1 (Sheet 4 of 7) are provided for shielding. On the east side, the stub wall provides shielding from radiation streaming from the east Steam Generator (SG 2). On the west side, the stub wall provides shielding from radiation streaming from the west steam generator (SG 1) and the pressurizer and pressurizer spray valves.

These different wall thicknesses were not previously identified in the UFSAR but have been incorporated into the dynamic, structural, and shielding calculations for the containment structural modules. They are being identified as a change in this LAR, because they were not previously specifically identified in the UFSAR. This change does not alter any design criteria, codes, standards or regulatory requirements. The modules with portions of the wall thickness outside the specified range in Figure 3.8.3-8 (Sheet 1) continue to satisfy the applicable requirements of AISC N690, ACI 349, and the supplemental requirements in UFSAR Subsection 3.8.3.5.3. The shielding provided by these walls continues to satisfy the radiation protection requirements and dose values as provided in UFSAR Sections 12.3 and 12.4, as these wall thicknesses were included in the shielding calculations forming the basis for the doses provided in the UFSAR. As a result, there is no impact on the doses already reported in the UFSAR, and, consequently, this change does not represent an ALARA concern. None of the figures in UFSAR Sections 1.2 and 12.3 are changed as a result of this activity. The seismic design and acceptance criteria, as provided in UFSAR Section 3.7, which are not altered by this change in wall thickness, continue to be applied to the walls with the changed wall thickness. Further, the wall thicknesses were considered in the global finite element analysis (FEA) models of the nuclear island, and providing the wall thickness clarification does not alter those models.

Structural Module Out-of-Plane Reinforcement

UFSAR Subsections 3.8.3.1.3, 3.8.3.5.3.5, and 3.8.3.5.8.1 are proposed to be changed to collectively state that steel plates, structural shapes, reinforcement bars, or tie bars between the faceplates of the CA01, CA02, and CA05 structural wall modules are added in some locations as additional structural elements in order to meet applicable code structural requirements. These added design features provide additional capacity compared with the current design that enables areas subject to local out-of-plane loads to satisfy the applicable requirements of AISC N690 and ACI 349. This change does not alter any design criteria, codes, standards or regulatory requirements but is needed to ensure that, at all locations, the structural wall modules are in compliance with the code requirements.

General Discussion Regarding All Changes

The proposed changes of the design details of the structural modules do not change the support, design, or operation of mechanical and fluid systems. The change of the detail design of the structural modules does not change the capacity, function, or response to anticipated transients or postulated accident conditions of any structure, system or component (SSC). There is no change to plant systems or the response of systems to postulated accident conditions. The proposed changes do not change the function, design, or operation of the containment vessel or the passive containment cooling system. The proposed changes do not affect the prevention or mitigation of abnormal events, e.g., accidents, anticipated operational occurrences, earthquakes, floods and turbine missiles, or their safety or design analyses. There is no change to the predicted radioactive releases due to normal operation or postulated accident conditions. The plant response to previously evaluated accidents or external events is not adversely affected, nor does the change described create any new accident precursors.

The proposed changes do not adversely affect any safety-related equipment, design code, design code allowable value, function or design analysis, nor do the proposed changes adversely affect any safety analysis input or result, or design/safety margin. The proposed changes do not interface with or affect safety-related equipment or a fission product barrier. No system or design function or equipment qualification would be adversely affected by the proposed changes. The changes do not result in a new failure mode, malfunction or sequence of events that could adversely affect a radioactive material barrier or safety-related equipment. The proposed changes do not allow for a new fission product release path, result in a new fission product barrier failure mode, or create a new sequence of events that would result in significant fuel cladding failures.

The proposed changes associated with this license amendment request include changes to detailed design of containment internal structures. The proposed changes do not affect the radiological source terms (i.e., amounts and types of radioactive materials released, their release rates and release durations) used in the accident analyses, thus, the consequences of accidents are not affected. These changes do not affect the containment, control, channeling, monitoring, processing or releasing of radioactive or non-radioactive materials. The location and design of penetrations and the permeability and waterproofing of the concrete in the exterior walls are not changed. The interface between the nuclear island and the external surrounding environment is not impacted by the proposed change. The types and quantities of expected effluents are not changed. No effluent release path is affected. The functionality of the design and operational features that are credited with controlling the release of effluents during plant operation is not diminished. Therefore, radioactive or non-radioactive material effluents are not affected. Plant radiation zones, controls required under 10 CFR Part 20, and expected amounts and types of radioactive materials are not affected by the proposed changes. Therefore, individual and cumulative radiation exposures do not change.

The proposed changes do not involve, nor interface with, any structure, system or component accident initiator or initiating sequence of events, and thus, the probabilities of the accidents evaluated in the UFSAR are not affected.

The proposed activity has no adverse effect on the ex-vessel severe accident. The overall design, geometry, and strength of the containment internal structures are not adversely affected. The design and material selection of the concrete floor beneath the reactor vessel is not altered. The response of the containment to a postulated reactor vessel failure, including direct containment heating, ex-vessel steam explosions, and core concrete interactions is not altered by the changes to the detail design of the structural modules. The design of the reactor vessel and the response of the reactor vessel to a postulated severe accident are not altered by the proposed changes.

The proposed activity has no impact on the Aircraft Impact Assessment. The changes described are internal to the structures and do not impact the design or response of the containment vessel and shield building. There is no change to protection of plant structures, systems, and components against aircraft impact provided by the design of the shield building. There is no change to the design of any of the key design features described in UFSAR Appendix 19F. The activity described does not change the overall design or construction of the shield building.

The change activity has no impact on the emergency plans or the physical security evaluation because there are no changes to the configuration of walls, doors, or access to the Nuclear Island.

Summary

The proposed changes would revise Tier 2, and associated Tier 1, COL Appendix C, and Tier 2* information in the UFSAR in regard to detailed design of containment internal structures. The proposed changes do not adversely affect the strength or response of the nuclear island structures.

The above proposed changes do not adversely affect any safety-related equipment or function, design function, radioactive material barrier or safety analysis.

4. Regulatory Evaluation

4.1 Applicable Regulatory Requirements/Criteria

10 CFR 52.98(f) requires NRC approval for any modification to, addition to, or deletion from the terms and conditions of a COL. This activity involves a departure from plant-specific Tier 1 information, and a corresponding change to COL Appendix C, Inspections, Tests, Analyses and Acceptance Criteria information; therefore, this activity requires a requested amendment to the COL. Accordingly, NRC approval is required prior to making the plant-specific changes in this license amendment request.

10 CFR 52, Appendix D, Section VIII.B.5.a allows an applicant or licensee who references this appendix to depart from Tier 2 information, without prior NRC approval, unless the proposed departure involves a change to or departure from Tier 1 information, Tier 2* information, or the Technical Specifications, or requires a license amendment under paragraphs B.5.b or B.5.c of the section. This change, which includes changes to design details for construction of the in-containment structural module wall attachment, faceplate thickness, wall thickness, and allows for the use of steel plates, structural shapes, reinforcement bars, or tie bars between the module faceplates, as needed to meet applicable code requirements, includes a Tier 2 departure which involves a revision to plant-specific Tier 1 information (and corresponding COL Appendix C information) and Tier 2* information, and thus requires NRC approval. Therefore, a license amendment request (LAR) (as supplied herein) is required.

10 CFR Part 50, Appendix A, General Design Criterion (GDC) 1 requires that structures be designed, fabricated, erected, constructed, tested, and inspected to quality standards commensurate with the importance of the safety functions to be performed. The proposed change does not change the criteria for the design, analysis, and construction of the in-containment structural module walls. The modules remain in conformance with the code requirements identified and supplemented in the UFSAR.

10 CFR Part 50, Appendix A, GDC 2 requires that structures withstand the effects of earthquakes and appropriate combinations of the effects of normal and accident conditions, including the effects of environmental loadings, such as earthquakes and other natural phenomena. The proposed changes have no adverse impact on the seismic motions to which the nuclear island structures are subjected and no adverse impact on the response of the nuclear island structures to seismic motions.

10 CFR Part 50, Appendix A, GDC 4 requires that systems, structures, and components can withstand the dynamic effects associated with missiles, pipe whipping, and discharging fluids, excluding dynamic effects associated with pipe ruptures, the probability of which is extremely low under conditions consistent with the design basis for the piping. The proposed changes do not adversely affect the configuration of the

walls and floors that provide separation between sources and potential targets. The proposed changes have no impact on the capability of the systems, structures, and components to withstand dynamic effects associated with missiles, pipe whipping, and discharging fluids as required by this criterion. The proposed changes do not change the requirements for anchoring safety-related components and supports to seismic Category I structures.

4.2 Precedent

No precedent is identified.

4.3 Significant Hazards Consideration

The requested amendment would revise the plant-specific Design Control Document (DCD) material incorporated into the Updated Final Safety Analysis Report (UFSAR), by revising the design details for the internal containment structural modules faceplate and wall thickness, the attachment of the containment structural modules to the base concrete, and providing for the use of steel plates, structural shapes, reinforcement bars, or tie bars between the module faceplates, where needed to meet applicable code requirements.

An evaluation to determine whether or not a significant hazards consideration is involved with the requested amendment was completed by focusing on the three standards set forth in 10CFR50.92, "Issuance of Amendment," as discussed below:

4.3.1 Does the requested amendment involve a significant increase in the probability or consequences of an accident previously evaluated?

Response: No

The design function of the internal containment structures is to provide support, protection, and separation for the seismic Category I mechanical and electrical equipment located in those structures. These structures are structurally designed to meet seismic Category I requirements as defined in Regulatory Guide 1.29.

The changes to the design details for the structural modules do not have an adverse impact on the response of the nuclear island structures to safe shutdown earthquake ground motions or loads due to anticipated transients or postulated accident conditions, nor do they change the seismic Category I classification. Evaluations have been performed which determined that the proposed changes do not have a significant impact on the calculated loads for the affected structural modules, or critical locations, and no significant impact on the global seismic model. The changes to the design details for the structural modules do not

impact the support, design, or operation of mechanical and fluid systems. There is no change to plant systems or the response of systems to postulated accident conditions. There is no change to the predicted radioactive releases due to postulated accident conditions. The plant response to previously evaluated accidents or external events is not adversely affected, nor does the change described create any new accident precursors.

Therefore, the requested amendment does not involve a significant increase in the probability or consequences of an accident previously evaluated.

4.3.2 Does the requested amendment create the possibility of a new or different kind of accident from any accident previously evaluated?

Response: No

The proposed changes are to revise design details for the internal containment structural modules. The changes do not change the design requirements of the nuclear island structures, nor do they change the seismic Category I classification. The changes to the design details for the internal containment structural modules do not change the design function, support, design, or operation of mechanical and fluid systems. The changes to the design details for the internal containment structural modules do not result in a new failure mechanism for the nuclear island structures or introduce any new accident precursors. As a result, the design function of the nuclear island structures is not adversely affected by the proposed change.

Therefore, the requested amendment does not create the possibility of a new or different kind of accident from any accident previously evaluated.

4.3.3 Does the requested amendment involve a significant reduction in a margin of safety?

Response: No

The requested amendment proposes changes to the structural details associated with the in-containment structural modules. The purpose of these changes is to ensure that the requirements contained in the applicable construction codes are met. As discussed in UFSAR, Section 3.8.3.5, "Design Procedures and Acceptance Criteria," the in-containment structural modules are designed in accordance with ACI 349 and AISC N690. Thus, the identification of additional structural module connection details, the increase in structural module faceplate and wall thicknesses, and the addition of additional reinforcement in specific areas are proposed to ensure that the codes of record, and the associated

margins contained therein, continue to be met as specified in the design basis. Structural and seismic analysis of the modified sections in accordance with the methodologies identified in the UFSAR has confirmed that the applicable requirements of ACI 349 and AISC N690 continue to be met for affected in-containment structural modules.

As a result, the proposed changes do not adversely affect any safety-related equipment or other design functions, design code compliance, design analysis, safety analysis input or result, or design/safety margin. No safety analysis or design basis acceptance limit/criterion is challenged or exceeded by the proposed changes. Therefore, the requested amendment does not involve a significant reduction in a margin of safety.

Based on the above, it is concluded that the requested amendment does not involve a significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and, accordingly, a finding of “no significant hazards consideration” is justified.

4.4 Conclusions

Based on the considerations discussed above, (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) such activities will be conducted in compliance with the Commission’s regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or to the health and safety of the public.

5. Environmental Considerations

The requested amendment revises the Updated Final Safety Analysis Report (UFSAR) Tier 2 and related plant-specific Design Control Document (DCD) Tier 1 (and corresponding COL Appendix C) and Tier 2* information by revising the design details for the internal containment structural modules.

A review has determined that the requested amendment would change a requirement with respect to installation or use of a facility component located within the restricted area, as defined in 10 CFR Part 20, or would change an inspection or surveillance requirement. However, facility construction and operation following implementation of the requested amendment does not involve (i) a significant hazards consideration, (ii) a significant change in the types or a significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation exposure. Accordingly, the requested amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9), in that:

(i) *There is no significant hazards consideration.*

As documented in Section 4.3, Significant Hazards Consideration Determination, of this license amendment request, an evaluation was completed to determine whether or not a significant hazards consideration is involved by focusing on the three standards set forth in 10 CFR 50.92, "Issuance of amendment." The Significant Hazards Consideration determined that (1) the requested amendment does not involve a significant increase in the probability or consequences of an accident previously evaluated; (2) the requested amendment does not create the possibility of a new or different kind of accident from any accident previously evaluated; and (3) the requested amendment does not involve a significant reduction in a margin of safety. Therefore, it is concluded that the requested amendment does not involve a significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and accordingly, a finding of "no significant hazards consideration" is justified.

(ii) *There is no significant change in the types or significant increase in the amounts of any effluents that may be released offsite.*

The requested amendment changes are unrelated to any aspect of plant construction or operation that would introduce any change to effluent types (e.g., effluents containing chemicals or biocides, sanitary system effluents, and other effluents), or affect any plant radiological or non-radiological effluent release quantities. Furthermore, the proposed changes do not affect any effluent release path or diminish the functionality of any design or operational features that are credited with controlling the release of effluents during plant operation. Therefore, it is concluded that the requested amendment does not involve a significant change in the types or a significant increase in the amounts of any effluents that may be released offsite.

(iii) *There is no significant increase in individual or cumulative occupational radiation exposure.*

The proposed changes to the design details for the internal containment structural modules do not adversely affect walls, floors, or other structures which provide shielding. Plant radiation zones are not affected, and there are no changes to the controls required under 10 CFR Part 20 that preclude a significant increase in occupational radiation exposure. Therefore, the requested amendment does not involve a significant increase in individual or cumulative occupational radiation exposure.

Based on the above review of the requested amendment, it has been determined that anticipated construction and operational impacts of the requested amendment do not involve (i) a significant hazards consideration, (ii) a significant change in the types or significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in the individual or cumulative occupational radiation exposure.

Accordingly, the requested amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the requested amendment.

Southern Nuclear Operating Company
Vogtle Electric Generating Plant (VEGP) Units 3 and 4

ND-14-0748

Enclosure 2

Request for Exemption
Regarding Containment Internal Structural Wall Module Design Details
(LAR-14-001)

1.0 Purpose

Southern Nuclear Operating Company (the Licensee) requests a permanent exemption from the provisions of 10 CFR 52, Appendix D, Section III.B, "Design Certification Rule for the AP1000 Design, Scope and Contents," to allow a departure from elements of the certification information in Tier 1 of the Generic DCD. The regulation, 10 CFR 52, Appendix D, Section III.B, requires an applicant or licensee referencing Appendix D to 10 CFR Part 52 to incorporate by reference and comply with the requirements of Appendix D, including certification information in DCD Tier 1. The Tier 1 information for which a plant-specific departure and exemption is being requested includes changes related to the design details of internal structural modules.

This request for exemption will apply the requirements of 10 CFR 52, Appendix D, Section VIII.A.4 to allow departures from generic Tier 1 information due to the proposed additions to the non-system based design descriptions and Inspections, Tests, Analyses, and Acceptance Criteria (ITAAC) as identified in Note 3 of Table 3.3-1, Definition of Wall Thicknesses for Nuclear Island Buildings, Turbine Building, and Annex Building.

Modular construction techniques are used extensively in the containment internal structures. Subassemblies are initially fabricated both offsite and onsite. Module assembly consists of combining the subassemblies into structural modules after which they are installed in the plant. Design finalization of the modules, fabrication experience in the shop environment, and the comparison of text and figures presented in the UFSAR have identified the need for proposed design changes to certain details of the structural modules identified in Tier 1 Table 3.3-1.

The use of thicker than nominal faceplates permits larger attachment or connection loads without the use of overlay plates. Structural modules CA01 and CA05 use thicker faceplates in some locations. In addition, overlay plates are attached in some locations to structural modules CA01, CA02, and CA05 to support equipment attachment. A thicker faceplate is also used in the area inside the northeast corner of the CA05 structural module where it transitions at lower elevations from 2'-0" (where the wall thickness is reduced to accommodate equipment) to 4'-6".

The additional thickness from the use of thicker faceplates or overlay plates is applicable to the outer surface of the module and, as a result, the overall module wall thickness is greater in these areas. This change affects dimensions specified in plant-specific DCD Tier 1, Table 3.3-1 and requires a revision to a note to the table.

2.0 Background

The Licensee is the holder of Combined License Nos. NPF-91 and NPF-92, which authorize construction and operation of two Westinghouse Electric Company AP1000 nuclear plants, named Vogtle Electric Generating Plant (VEGP) Units 3 and 4, respectively.

The containment internal structures are those concrete and steel structures inside (not part of) the containment pressure boundary that support the reactor coolant system components and

related piping systems and equipment. The walls of these modules are part of the containment internal structures and provide support for floors and other steel structures, component and piping supports, and other attachments. Structural wall modules are used for the primary shield wall around the reactor vessel, the wall between the vertical access and the Chemical and Volume Control System (CVS) room, secondary shield walls around the steam generators and pressurizer, the east side of the in-containment refueling water storage tank, and the refueling cavity. Structural wall modules consist of steel faceplates connected by steel trusses. Following installation of the structural module within the containment building, the hollow wall structures are filled with concrete, forming a portion of the structural walls of the containment internal structures.

The containment internal structural module is a seismic Category I structure and is designed for dead, live, thermal, pressure, safe shutdown earthquake, and loads due to postulated pipe breaks.

Supports for piping, electrical raceways, equipment, and components are attached to the module structures by welding the supports directly to the structural module faceplate, or to overlay plates welded to the structural module faceplates. The loads from the supports are transferred through the overlay plates, where used; to the structural wall module faceplates and shear studs or backup structure elements into the concrete. The UFSAR identifies the nominal thickness of the structural wall module faceplates as 0.5 inch in text, figures, and tables. In some cases the support loads require more capacity than that provided by the 0.5 inch faceplates and shear studs. Where faceplates thicker than the nominal 0.5 inches or overlay plates are used in the construction of the structural wall modules, the wall thicknesses in the area of the thicker faceplates and overlay plates are greater than indicated in Tier 1 Table 3.3.1 by the amount of faceplate and/or overlay plate thickness increase over the nominal 0.5 inches. Therefore, it is proposed to add following two sentences to Note 3 of Tier 1 Table 3.3.1:

“Where faceplates with a nominal thickness of 0.5 inches are used in the construction of the wall modules, the wall thicknesses in this column apply. Where faceplates thicker than the nominal 0.5 inches or overlay plates are used in the construction of the structural wall modules, the wall thicknesses in the area of the thicker faceplates and overlay plates are greater than indicated in this column by the amount of faceplate and/or overlay plate thickness increase over the nominal 0.5 inches”.

3.0 Technical Justification of Acceptability

The use of thicker than nominal faceplates permits larger attachment or connection loads without the use of overlay plates or additional backup structures. In addition, heavy sections or structural shapes may be welded to the faceplates during module fabrication. The use of thicker faceplates in these areas facilitates the fabrication of the modules. The additional thickness is applied to the outer surface of the module and, as a result, the overall module wall thickness is greater in these areas. The design of the modules with the subject variances is in conformance with design and analysis requirements for the modules identified in the UFSAR. The modules with portions of the faceplates thicker than 0.5 inch satisfy the applicable requirements of AISC N690, ACI 349 and the supplemental requirements in UFSAR Subsection 3.8.3.5.3. The thicker faceplates do not have an adverse effect on the seismic or structural analysis of the structural modules based on evaluations supporting the design change. Because the thicker faceplates are used for local attachment loads and connections and are not required for seismic loads on the structure, the spacing of the shear studs is not changed.

Where overlay plates are used, they function as load attachment and distribution elements. The overlay plates do not affect the structural wall module stiffness used in the global model and do not participate in resisting the internal wall module forces from the global model, including the global seismic analysis.

Additional detail for justification for this exemption is provided in Section 3 of the accompanying License Amendment Request in Enclosure 1.

4.0 Justification of Exemption

10 CFR Part 52, Appendix D, Section VIII.A.4 and 10 CFR 52.63(b)(1) govern the issuance of exemptions from elements of the certified design information for AP1000 nuclear power plants. Because the Licensee has identified changes to the Tier 1 information related to the Tier 2 departure discussed in Enclosure 1 of the accompanying License Amendment Request, an exemption from the certified design information in Tier 1 is needed.

10 CFR Part 52, Appendix D, and 10 CFR §§ 50.12, 52.7, and 52.63 state that the NRC may grant exemptions from the requirements of the regulations provided six conditions are met: 1) the exemption is authorized by law [§50.12(a)(1)]; 2) the exemption will not present an undue risk to the health and safety of the public [§50.12(a)(1)]; 3) the exemption is consistent with the common defense and security [§50.12(a)(1)]; 4) special circumstances are present [§50.12(a)(2)]; 5) the special circumstances outweigh any decrease in safety that may result from the reduction in standardization caused by the exemption [§52.63(b)(1)]; and 6) the design change will not result in a significant decrease in the level of safety [Part 52, App. D, VIII.A.4].

The requested exemption satisfies the criteria for granting specific exemptions, as described below.

1. This exemption is authorized by law

The NRC has authority under 10 CFR §§ 50.12, 52.7, and 52.63 to grant exemptions from the requirements of NRC regulations. Specifically, 10 CFR §§50.12 and 52.7 state that the NRC may grant exemptions from the requirements of 10 CFR Part 52 upon a proper showing. No law exists that would preclude the changes covered by this exemption request. Additionally, granting of the proposed exemption does not result in a violation of the Atomic Energy Act of 1954, as amended, or the Commission's regulations.

Accordingly, this requested exemption is "authorized by law," as required by 10 CFR §50.12(a)(1).

2. This exemption will not present an undue risk to the health and safety of the public

The proposed exemption from the requirements of 10 CFR 52, Appendix D, Section III.B would allow changes to elements of the plant-specific Tier 1 DCD to depart from the AP1000 certified (Tier 1) design information. The plant-specific Tier 1 DCD will continue to reflect the approved licensing basis for the Licensee, and will maintain a consistent level of detail with that which is currently provided elsewhere in Tier 1 of the plant-specific DCD. The changes to the design details for the structural wall modules do not have an adverse impact on the response of the nuclear island structures to safe shutdown earthquake ground motions or loads due to anticipated transients or postulated accident conditions, nor do they change the seismic Category I classification. The changes to the design details for the structural wall modules do not impact the support, design, or operation of mechanical and fluid systems. There is no change to plant systems or the response of systems to postulated accident conditions. There is no change to the predicted radioactive releases due to postulated accident conditions. The plant response to previously evaluated accidents or external events is not adversely affected, and the change described does not create any new accident precursors. Therefore, no adverse safety impact that would present any additional risk to the health and safety is present. The affected Design Description in the plant-specific Tier 1 DCD will also continue to provide the detail necessary to support the performance of the associated ITAAC.

Therefore, the requested exemption from 10 CFR 52, Appendix D, Section III.B would not present an undue risk to the health and safety of the public.

3. The exemption is consistent with the common defense and security

The exemption from the requirements of 10 CFR 52, Appendix D, Section III.B would change elements of the plant-specific Tier 1 DCD by departing from the AP1000 certified (Tier 1) design information. The exemption does not alter the design, function, or operation of any structures or plant equipment that is necessary to maintain a safe and secure status of the plant. The proposed exemption has no impact on plant security or safeguards procedures.

Therefore, the requested exemption is consistent with the common defense and security.

4. Special circumstances are present

10 CFR 50.12(a)(2) list six “special circumstances” for which an exemption may be granted. Pursuant to the regulation, it is necessary for one of these special circumstances to be present in order for the NRC to consider granting an exemption request. The requested exemption meets the special circumstances of 10 CFR 50.12(a)(2)(ii). That subsection defines special circumstances as when “Application of the regulation in the particular circumstances would not serve the underlying purpose of the rule or is not necessary to achieve the underlying purpose of the rule.”

The rule under consideration in this request for exemption is 10 CFR 52, Appendix D, Section III.B, which requires that a licensee referencing the AP1000 Design Certification Rule (10 CFR Part 52, Appendix D) shall incorporate by reference and comply with the requirements of Appendix D, including Tier 1 information. The VEGP Units 3 and 4 COLs reference the AP1000 Design Certification Rule and incorporate by reference the requirements of 10 CFR Part 52, Appendix D, including Tier 1 information. The underlying purpose of Appendix D, Section III.B is to describe and define the scope and contents of the AP1000 design certification, and to require compliance with the design certification information in Appendix D.

The proposed changes to the design details for the structural wall modules maintain the design margins of the internal containment structures. This change does not impact the ability of any structures, systems, or components to perform their functions or negatively impact safety. Accordingly, this exemption from the certification information will enable the Licensee to safely construct and operate the AP1000 facility consistent with the design certified by the NRC in 10 CFR 52, Appendix D.

Therefore, special circumstances are present, because application of the current generic certified design information in Tier 1 as required by 10 CFR Part 52,

Appendix D, Section III.B, in the particular circumstances discussed in this request is not necessary to achieve the underlying purpose of the rule.

5. The special circumstances outweigh any decrease in safety that may result from the reduction in standardization caused by the exemption

Based on the nature of the changes to the plant-specific Tier 1 information and the understanding that these changes support the design function of the internal containment structures, it is expected that this exemption will be requested by other AP1000 licensees and applicants. However, a review of the reduction in standardization resulting the departure from the standard DCD determined that even if other AP1000 licensees and applicants do not request this same departure, the special circumstances will continue to outweigh any decrease in safety from the reduction in standardization because the key design functions of the structures associated with this request will continue to be maintained. Furthermore, the justification provided in this exemption request and the associated marked-up tables demonstrate that the change from the standard information provided in the generic AP1000 DCD is offset by the special circumstances identified above.

Therefore, the special circumstances associated with the requested exemption outweigh any decrease in safety that may result from the reduction in standardization caused by the exemption. In fact, as described in item 6 below, the exemption will result in no reduction in the level of safety.

6. The design change will not result in a significant decrease in the level of safety.

The exemption revises the plant-specific DCD Tier 1 information by altering the design details for the internal containment structural wall modules. The changes to the design details for the internal containment structural wall modules do not change the design requirements of the nuclear island structures. Because these functions continue to be met, there is no reduction in the level of safety.

5.0 Risk Assessment

A risk assessment was determined to be not applicable to address the acceptability of this request.

6.0 Precedent

No precedent for this request is identified.

7.0 Environmental Consideration

A review has determined that the proposed exemption would change a requirement with respect to installation or use of a facility component located within the restricted area, as defined in 10 CFR 20, or would change an inspection or surveillance requirement. However, the proposed exemption does not involve (i) a significant hazards consideration, (ii) a significant change in the types or a significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation exposure. Specific justification is provided in Section 5 of the corresponding License Amendment Request in Enclosure 1. Accordingly, the proposed exemption meets the eligibility criterion for categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the proposed exemption.

8.0 Conclusion

The proposed changes to Tier 1 are necessary to revise Note 3 in plant-specific Tier 1 Table 3.3-1 to add that where thicker faceplates or overlay plates are used in the construction of the structural wall modules, the wall thicknesses in the area of the thicker faceplates and overlay plates are larger than indicated in the table. The exemption request meets the requirements of 10 CFR 52.63, *"Finality of design certifications,"* 10 CFR 52.7, *"Specific exemptions,"* 10 CFR 50.12, *"Specific exemptions,"* 10 CFR 51.22, *"Criteria for categorical exclusion; identification of licensing and regulatory actions eligible for categorical exclusion or otherwise not requiring environmental review,"* and 10 CFR 52 Appendix D, *"Design Certification Rule for the AP1000."* Specifically, the exemption request meets the criteria of 10 CFR 50.12(a)(1) in that the request is authorized by law, presents no undue risk to public health and safety, and is consistent with the common defense and security. Furthermore, approval of this request does not result in a significant decrease in the level of safety, presents special circumstances, does not present a significant decrease in safety as a result of a reduction in standardization, and meets the eligibility requirements for categorical exclusion.

Southern Nuclear Operating Company
Vogtle Electric Generating Plant (VEGP) Units 3 and 4

ND-14-0748

Enclosure 3

Proposed Changes to Licensing Basis Documents
(LAR-14-001)

Tier 1 Changes

Plant-Specific Tier 1 Table 3.3-1 and corresponding COL Appendix C Table 3.3-1, Definition of Wall Thicknesses for Nuclear Island Buildings, Turbine Building, and Annex Building – Revise to add addition to Note 3 as shown below:

Table 3.3-1 Definition of Wall Thicknesses for Nuclear Island Buildings, Turbine Building, and Annex Building⁽¹⁾				
Wall or Section Description	Column Lines⁽⁷⁾	Floor Elevation or Elevation Range⁽⁷⁾⁽⁸⁾	Concrete Thickness⁽²⁾⁽³⁾⁽⁴⁾⁽⁵⁾⁽⁹⁾	Applicable Radiation Shielding Wall (Yes/No)
Containment Building Internal Structure				
Shield Wall between Reactor Vessel Cavity and RCDT Room	E-W wall parallel with column line 7 (Inside face is 3'-0" north of column line 7. Width of wall section with stated thickness is defined by inside wall of reactor vessel cavity.)	From 71'-6" to 83'-0"	3'-0"	Yes
West Reactor Vessel Cavity Wall	N-S wall parallel with column line N (Width of wall section with stated thickness is defined by inside wall of reactor vessel cavity).	From 83'-0" to 98'-0"	7'-6"	Yes
North Reactor Vessel Cavity Wall	E-W wall parallel with column line 7 (Width of wall section with stated thickness is defined by inside wall of reactor vessel cavity).	From 83'-0" to 98'-0"	9'-0"	Yes

1. The column lines and floor elevations are identified and included on Figures 3.3-1 through 3.3-13.
2. These wall (and floor) thicknesses have a construction tolerance of ± 1 inch, except for exterior walls below grade where the tolerance is +12 inches, -1 inch. These tolerances are not applicable to the nuclear island basemat.
3. For walls that are part of structural modules, the concrete thickness also includes the steel face plates. [Where faceplates with a nominal thickness of 0.5 inches are used in the construction of the wall modules, the wall thicknesses in this column apply. Where faceplates thicker than the nominal 0.5 inches or overlay plates are used in the construction of the structural wall modules, the wall thicknesses in the area of the thicker faceplates and overlay plates are greater than indicated in this column by the amount of faceplate and/or overlay plate thickness increase over the nominal 0.5 inches.](#)
4. For floors with steel surface plates, the concrete thickness also includes the plate thickness.
5. Where a wall (or a floor) has openings, the concrete thickness does not apply at the opening.
6. The elevation ranges for the shield building items are rounded to the nearest inch.
7. The Wall or Section Description, Column Line information, and Floor Elevation or Elevation Ranges are provided as reference points to define the general location. The concrete thickness of an item intersecting other walls, roofs or floors at a designated location (e.g., column line) is not intended to be measured to the stated column line, but only to the point where the intersection occurs.
8. Where applicable, the upper wall portions extend to their associated roofs, which may vary in elevation, e.g., sloped roofs.
9. From one wall/floor section to another, the concrete thickness transitions from one thickness to another, consistent with the configurations in Figures 3.3-1 through 3.3-14.

Tier 1 Material

Tier 2* Changes and Associated Tier 2 Changes

UFSAR Subsection 3.8.3.1.3, Structural Wall Modules - Revise second paragraph to read as shown below:

Structural wall modules consist of steel faceplates connected by steel trusses. The primary purpose of the trusses is to stiffen and hold together the faceplates during handling, erection, and concrete placement. ~~At corner locations, the trusses are replaced with diaphragms. These diaphragms have similar purpose as the trusses and are used to fabricate the modules in the corners. The nominal spacing of the trusses is 30 inches.~~ The nominal thickness of the steel faceplates is 0.5 inch. ~~The nominal spacing of the trusses is 30 inches. [In some locations within the CA01 and CA05 structural wall modules, the faceplate thickness is greater than the nominal thickness to support local demand. For areas subject to these high out-of-plane loads, the faceplate thickness may be increased up to a thickness of 1.5 inches. To support loading associated with the steam generator lateral supports, the faceplate thickness is increased to 3.0 inches.]*~~ At corner locations, the trusses are replaced with diaphragms. These diaphragms have similar purpose as the trusses and are used to fabricate the modules in the corners. The detail design of the trusses and the design of the diaphragms shown in Figure 3.8.3-8, Sheet 1 are representative. Details such as the location, type, and size of welds and length and end configuration of channels used to assemble the internal trusses may vary in the design implemented from that shown in the figure. The size, shape, and location of welds, structural shapes, and other elements used to fabricate and stiffen the diaphragms and connect them to the faceplates and the location, size, and shape of the access opening may vary from that shown in the figure. Steel plates, structural shapes, reinforcement bars, or tie bars are installed between the structural wall module faceplates and embedded in the concrete as additional structural elements that support out-of-plane loads and structural integrity. This reinforcement is designed to the applicable requirements of ACI 349 and AISC N690. Each module has different characteristics that require the detailed design to be specific for that module and loading and interferences within that module. Shear studs are welded to the inside faces of the steel faceplates. Face plates are welded to adjacent plates with full penetration welds so that the weld is at least as strong as the plate. The full penetration welds are identified in Figure 3.8.3-8, Sheet 1 with the weld symbol that includes the notation CJP for complete joint penetration. Plates on each face of the wall module extend down to the elevation of the adjacent floor. Where the floors in the rooms on each side of the wall module are at different elevations, one of the plates at the bottom of the module extends further down than the other. This single-sided configuration is designated on Figure 3.8.3-1 as “CA Structure Module with Single Surface Plate” for lower elevations in containment. A typical example with the single-sided configuration at the bottom of the module is shown in Figure 3.8.3-8. Some modules or portions of modules do not have the single-sided configuration at the bottom of the module. The module functions as a wall above the upper floor level (elevation 103'-0" in Figure 3.8.3-8). The single plate below this elevation is designed to transfer the reactions at the base of the wall into the basemat. This plate also acts as face reinforcement for the basemat. Basemat reinforcement dowels are provided at the bottom of the single plate as shown in Figure 3.8.3-8.

UFSAR Subsection 3.8.3.1.3, Structural Wall Modules – Add new paragraph after second paragraph as shown below:

[Shear studs, structural shapes, or horizontally oriented reinforcement bars mechanically connected or welded to the module faceplates may be used to anchor the CA01 and CA05 structural modules to the base concrete. Shear lugs designed in accordance with ACI 349 may also be used to transfer loads at the connection. Where horizontal bars are attached perpendicular to the faceplate, they function like shear studs to provide for faceplate development. Where additional reinforcement is required for detailing base concrete at module walls with offsets in faceplate elevation, the reinforcement is developed in accordance with ACI 349. Where horizontal bars are attached to the edge of a faceplate at an elevation offset, the bars are designed for a combination of applicable demand and developed in accordance with ACI 349.]*

UFSAR Subsection 3.8.3.1.3, Structural Wall Modules - Revise third paragraph to read as shown below:

[The information in Figure 3.8.3-8, Sheet 1 that is considered to be Tier 2* information is the design spacing of the faceplates, trusses, channels in the trusses and the minimum size and design spacing of the headed studs for the modular wall in the containment internal structure in locations away from openings or penetrations in the walls. Local variation in the design of the trusses and spacing of the trusses and shear studs may be required to address internal obstructions and accessibility for fabrication and inspection. As shown in Figure 3.8.3-1, Sheets 2 and 4, some CA01 and CA05 wall thicknesses are different than the wall thicknesses listed in Figure 3.8.3-8. Wall thickness increase due to local faceplate thickness increase is below the level of detail depicted in Figure 3.8.3-1. The module wall thicknesses are determined based on radiation protection considerations and are designed to the applicable requirements of AISC N690 and ACI 349. The use of full penetration welds to connect the faceplates of the modules is also considered to be Tier 2* information. The information in Figure 3.8.3-8, Sheet 2 that is considered to be Tier 2* information is the use of mechanical connectors and the development length requirement for the mechanical connectors.]* The detail design of the mechanical connectors is governed by AISC N690 and ACI 349, and the representative design shown is not considered to be Tier 2* information. The material, size, and configuration of plates, size and type of welds, method of attaching the connectors to the modules, and size of deformed bars anchoring the embedments into the concrete may be different for each module and may vary from that shown in the figure provided that the embedment design satisfies the AISC N690 and ACI 349 codes. Sheet 3 of Figure 3.8.3-8 shows a wall of the IRWST that is a steel structure anchored in concrete and is not a concrete filled module. [The information in Figure 3.8.3-8, Sheet 3 that is considered to be Tier 2* information is the plate thickness for the IRWST wall, the use, spacing, and size of angles and tees to stiffen the wall, the number, size, and use of studs provided to anchor the module, and the number, size, vertical spacing, and development length of the deformed bars provided to connect the module to the mass concrete. The design implemented in fabrication and construction drawings and instructions will have the design shown, an equal design, or a better design for the key structural elements.]*

UFSAR Subsection 3.8.3.1.3, Structural Wall Modules - Revise fourth paragraph to read as shown below:

The structural wall modules are anchored to the concrete base by reinforcing steel dowels or other types of connections mechanically connected or welded to the modules and embedded in the reinforced concrete below or adjacent to the module. [The reinforcing steel used to anchor the modules to the concrete may be oriented horizontally or vertically and has a development that satisfies the requirements of ACI 349.]* After erection, concrete is placed between the faceplates. Typical design details of the structural modules are shown in Figures 3.8.3-2 and 3.8.3-8. Shear-friction reinforcement may be included in construction joints at the transition between the base concrete and concrete in the modules and is not part of the mechanical connection between the module and base concrete described above. The shear-friction reinforcement at the construction joints is spaced and sized in accordance with ACI 349 requirements and is not connected to the module faceplates.

UFSAR Subsection 3.8.3.5.3, Structural Wall Modules - Revise third paragraph to read as shown below:

Figure 3.8.3-8 shows the typical design details of the structural modules, typical configuration of the wall modules, typical anchorages of the wall modules to the reinforced base concrete, and connections between adjacent modules. Variations in the module design details and design elements used are described in the Figure 3.8.3-8 notes and in subsections 3.8.3.1 and 3.8.3.1.3. Concrete-filled structural wall modules are designed as reinforced concrete structures in accordance with the requirements of ACI-349, as supplemented in the following paragraphs. The faceplates are considered as the reinforcing steel, bonded to the concrete by headed studs. The application of ACI-349 and the supplemental requirements are supported by the behavior studies described in Subsection 3.8.3.4.1. The steel plate modules are anchored to the reinforced concrete basemat by mechanical connections welded to the steel plate. Loads are transferred directly from the faceplates to the base concrete using reinforcing bars, structural shapes, and shear studs connected directly to the modules with mechanical connectors, ~~or~~ welds. The reinforcing steel used to anchor the modules to the concrete has a development that satisfies the requirements of ACI 349. The design of the surface plate, base plate, and vertical stiffeners is checked by finite element analysis. The design of critical sections is described in Subsection 3.8.3.5.8.

UFSAR Subsection 3.8.3.5.3.5, Design of Trusses - Revise to read as shown below:

The trusses provide a structural framework for the modules, maintain the separation between the faceplates, support the modules during transportation and erection, and act as "form ties" between the faceplates when concrete is being placed. After the concrete has cured, the trusses are not required to contribute to the strength or stiffness of the completed modules. However, they do provide additional shear capacity between the steel plates and concrete as well as additional strength similar to that provided by stirrups in reinforced concrete. As described in subsection 3.8.3.1.3, additional structural capacity is provided between the module faceplates in some areas to support localized loads. The trusses, connecting steel plates, structural shapes, reinforcement bars, and tie bars are designed according to the applicable requirements of AISC ~~—~~ N690 and ACI 349.

UFSAR Subsection 3.8.3.5.8.1, Structural Wall Modules - Revise paragraphs after the bulleted list to read as shown below:

The thicknesses and locations of these walls which are part of the boundary of the in-containment refueling water storage tank are shown in Figure 3.8.3-18. They are the portions of the structural wall modules experiencing the largest demand. The structural configuration and typical details are shown in Figures 3.8.3-1, 3.8.3-2, 3.8.3-8, 3.8.3-14, 3.8.3-15, and 3.8.3-17. The details shown in Figure 3.8.3-17 are representative of connections between floors in containment and walls constructed using steel plate concrete composite construction. Plate thickness, structural shape size, and reinforcement provided may be increased ~~due to~~ locally ~~higher loads~~. The design implemented in fabrication and construction drawings and instructions may have alternative structural shapes or reinforcement arrangements if they provide equal or better load capacity. As described in Subsection 3.8.3.1.3, steel plates, structural shapes, reinforcement bars, or tie bars are used in other locations between the structural module faceplates to address local out-of-plane loads.] The structural analyses are described in Subsection 3.8.3.4 and summarized in Table 3.8.3-2. The design procedures are described in Subsection 3.8.3.5.3.*

UFSAR Subsection 3.8.3.8, Construction Inspection - Revise to add the following after the bulleted list as shown below:

- Structural margin
- Ease of construction
- Construction loads
- Use of standard thicknesses

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness.]*

UFSAR Table 3.8.3-3, Definition of Critical Locations and Thicknesses for Containment Internal Structures - Revise as described below and as shown in UFSAR Table 3.8.3-3 on page 7 of 18:

- In the column titled "Thickness of Surface Plates Provided", remove the leading period for the 0.01 values in all three rows to fix a typographical error.
- Remove the word "maximum" in the title of the fifth column. These values do not represent maximum values, because they are less than the nominal values provided in the sixth column of the table.
- Add a footnote (4) to the title of the table.
- Add the following Footnote 4:

[4 As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness.]*

Table 3.8.3-3
Definition of Critical Locations and Thicknesses for Containment Internal Structures⁽¹⁾⁽⁴⁾

Wall Description (see detail in subsection 3.8.3.5.8.1)	Applicable Column Lines	Applicable Elevation Range	[Concrete Thickness ⁽²⁾]*	Required Thickness of Surface Plates (inches) ⁽³⁾ (Maximum)	[Thickness of Surface Plates Provided (inches)]*
Containment Structures					
Module Wall 1	West wall of refueling cavity	Wall separating IRWST and refueling cavity from elevation 103' to 135'-3"	[4'-0" concrete-filled structural wall module with 0.5-in.-thick steel plate on inside and outside of wall]*	0.30	[0.5 - -0.01 +0.1]*
Module Wall 2	South wall of west steam generator cavity	Wall separating IRWST and west steam generator cavity from elevation 103' to 135'-3"	[2'-6" concrete-filled structural wall module with 0.5-in.-thick steel plate on inside and outside of wall]*	0.44	[0.5 - -0.01 +0.1]*
CA02 Module Wall	North east boundary wall of IRWST	Wall separating IRWST and maintenance floor from elevation 103' to 135'-3"	[2'-6" concrete-filled structural wall module with 0.5-in.-thick steel plate on inside and outside of wall]*	0.37	[0.5 - -0.01 +0.1]*

Notes:

1. The applicable column lines and elevation levels are identified and included in Figures 1.2-9, 3.7.2-12 (sheets 1 through 12), 3.7.2-19 (sheets 1 through 3) and on Table 1.2-1.
2. The concrete thickness includes the steel face plates. Thickness greater than 3'-0" have a construction tolerance of +1", -3/4". Thickness less than or equal to 3'-0" have a construction tolerance of +1/2", -3/8".*
3. These plate thicknesses represent the thickness required for operating and design basis loads except for designed openings or penetrations. These values apply for each face of the applicable wall unless specifically indicated on the table. For load combinations with thermal loads, the evaluation is performed as described in Subsection 3.8.3.5.3.4.
4. Portions of the module faceplates may have a design thickness ranging from the nominal 0.5 inches to 1.5 inches to provide additional strength for localized loads due to attachments or connections beyond the operating and design basis loads addressed in Note 3.*

**UFSAR Table 3.8.3-4, (Sheet 1 of 3), Design Summary of West Wall of Refueling Canal Design Loads, Load Combinations, and Comparison to Acceptance Criteria Mid-Span at Mid- Height -
Revise the notes as shown below:**

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 101870

Plate thickness required for load combinations excluding thermal: 0.100 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 23.37 ksi

[Yield stress at temperature: 65.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 23.37 ksi

Allowable stress intensity range for load combinations including thermal: 130.0 ksi (Minimum)

UFSAR Table 3.8.3-4, (Sheet 2 of 3), Design Summary of West Wall of Refueling Canal Design Loads, Load Combinations, and Comparison to Acceptance Criteria Mid-Span at Base - Revise the notes as shown below:

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 101788

Plate thickness required for load combinations excluding thermal: 0.100 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 28.0 ksi

[Yield stress at temperature: 65.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 28.0 ksi

Allowable stress intensity range for load combinations including thermal: 130.0 ksi (Minimum)

**UFSAR Table 3.8.3-4, (Sheet 3 of 3), Design Summary of West Wall of Refueling Canal Design Loads, Load Combinations, and Comparison to Acceptance Criteria North End Bottom Corner -
Revise the notes as shown below:**

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 101794

Plate thickness required for load combinations excluding thermal: 0.30 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 28.1 ksi

[Yield stress at temperature: 65.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 35.26 ksi

Allowable stress intensity range for load combinations including thermal: 130.0 ksi (Minimum)

**UFSAR Table 3.8.3-5, (Sheet 1 of 3), Design Summary of South Wall of Steam Generator
Compartment Design Loads, Load Combinations, and Comparison to Acceptance Criteria Mid-Span
at Mid- Height - Revise the notes as shown below:**

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 104228

Plate thickness required for load combinations excluding thermal: 0.100 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 23.0 ksi

[Yield stress at temperature: 36.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 23.0 ksi

Allowable stress intensity range for load combinations including thermal: 72.0 ksi (Minimum)

**UFSAR Table 3.8.3-5, (Sheet 2 of 3), Design Summary of South Wall of Steam Generator
Compartment Design Loads, Load Combinations, and Comparison to Acceptance Criteria Mid-Span
at Base - Revise the notes as shown below:**

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 101943

Plate thickness required for load combinations excluding thermal: 0.100 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 25.7 ksi

[Yield stress at temperature: 36.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 25.7 ksi

Allowable stress intensity range for load combinations including thermal: 72.0 ksi (Minimum)

**UFSAR Table 3.8.3-5, (Sheet 3 of 3), Design Summary of South Wall of Steam Generator
Compartment Design Loads, Load Combinations, and Comparison to Acceptance Criteria West End
Bottom Corner - Revise the notes as shown below:**

Notes:

x-direction is horizontal; y-direction is vertical.

Element number 101933

Plate thickness required for load combinations excluding thermal: 0.100 inches (Maximum)

[Plate thickness provided: 0.50 inches -0.01 +0.10

[As described in subsection 3.8.3.1.3, some CA01 and CA05 module wall faceplate thicknesses are greater than the 0.5-inch nominal faceplate thickness. \]*](#)

Maximum principal stress for load combinations including thermal: 43.1 ksi

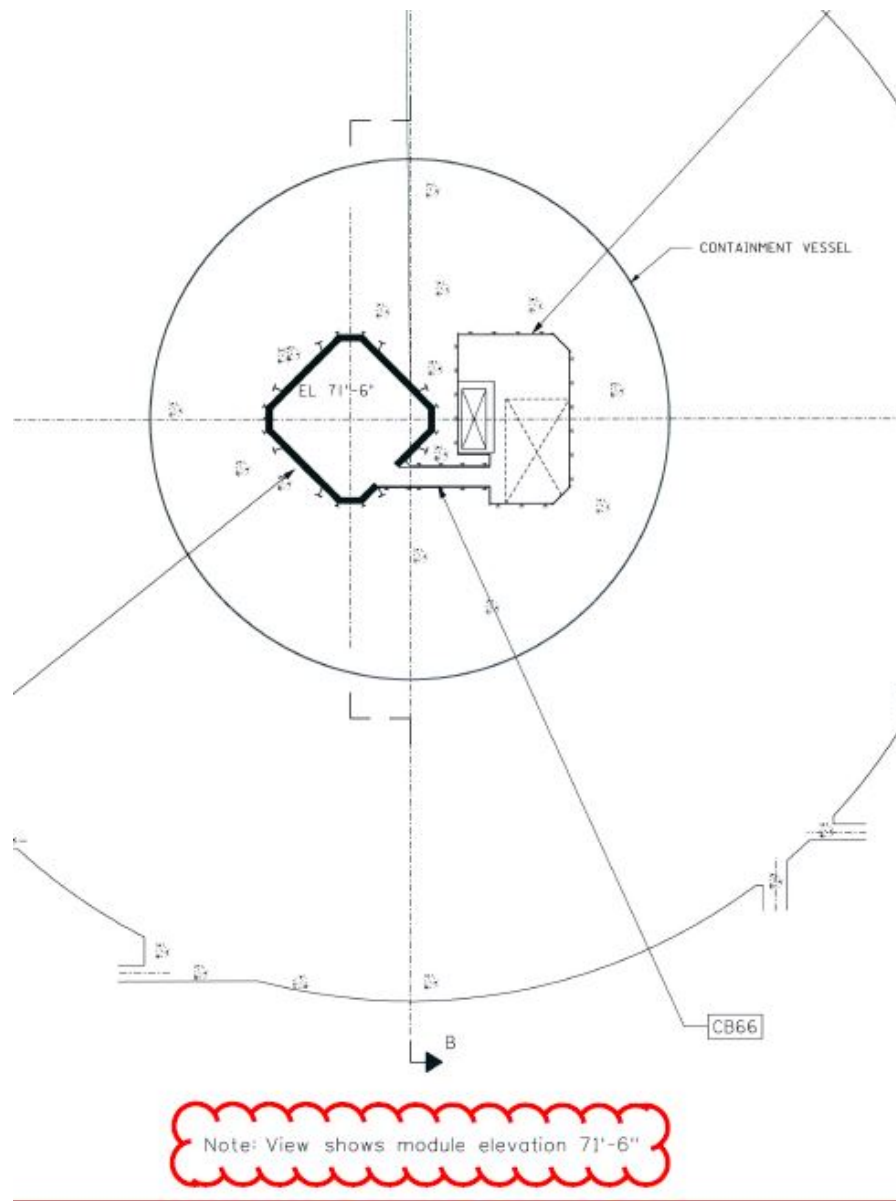
[Yield stress at temperature: 36.0 ksi (Minimum)]*

Maximum stress intensity range for load combinations including thermal: 52.6 ksi

Allowable stress intensity range for load combinations including thermal: 72.0 ksi (Minimum)

UFSAR Figure 3.8.3-1, (Sheet 1 of 7) Structural Modules in Containment Internal Structure – Revise to add a note stating “Note: View shows module elevation 71’-6”” as shown in Figure 1 below:

Figure 1



ND-14-0748

Enclosure 3

Proposed Changes to Licensing Basis Documents (LAR-14-001)

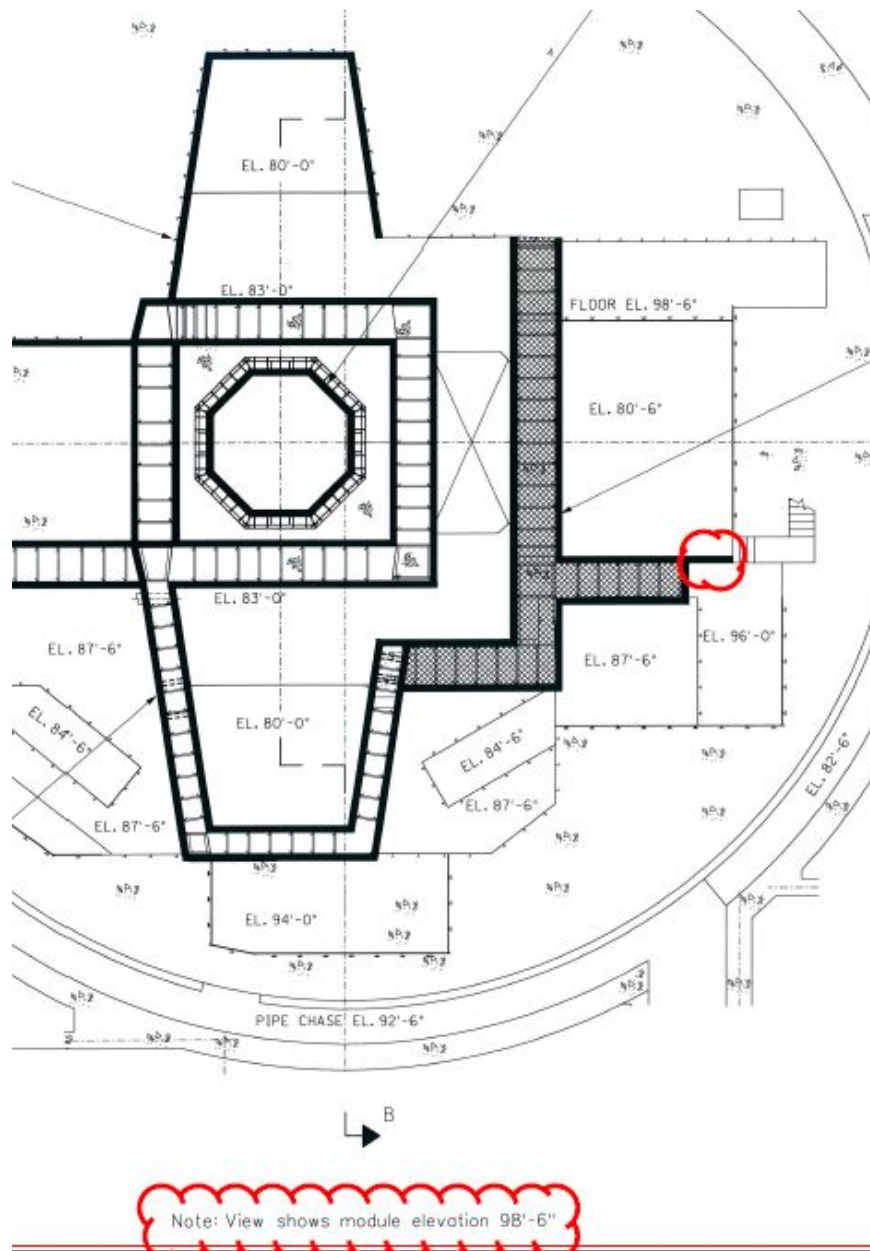
UFSAR Figure 3.8.3-1, (Sheet 2 of 7) Structural Modules in Containment Internal Structures – Revise to add a note stating “Note: View shows module elevation 87'-6”” as shown in Figure 2 below:

Figure 2

[This figure contains Security Related Information. See Enclosure 4]

UFSAR Figure 3.8.3-1, (Sheet 3 of 7) Structural Modules in Containment Internal Structures – Revise to add a note stating “Note: View shows module elevation 98’-6”” as shown in Figure 3 below:

Figure 3



ND-14-0748

Enclosure 3

Proposed Changes to Licensing Basis Documents (LAR-14-001)

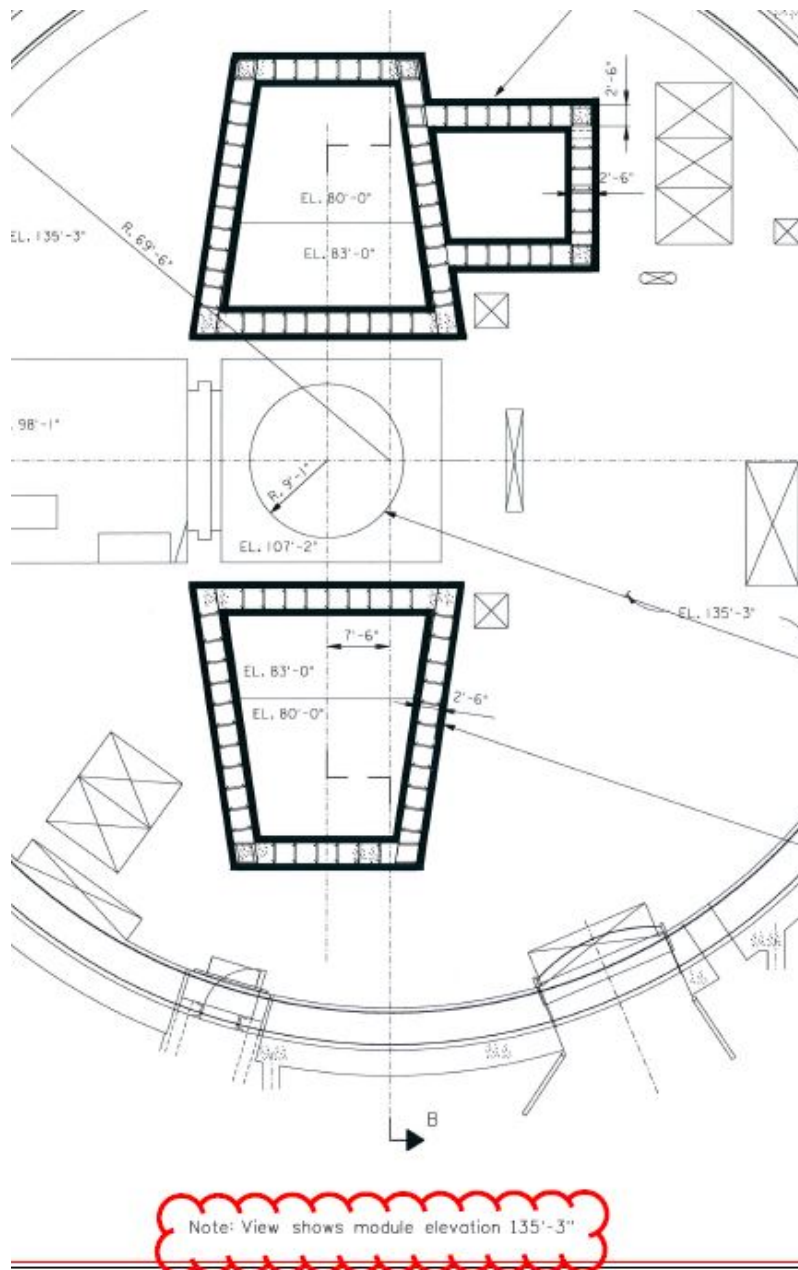
UFSAR Figure 3.8.3-1, (Sheet 4 of 7) Structural Modules in Containment Internal Structures – Revise to add a note stating “Note: View shows module elevation 107'-2”” as shown in Figure 4 below:

Figure 4

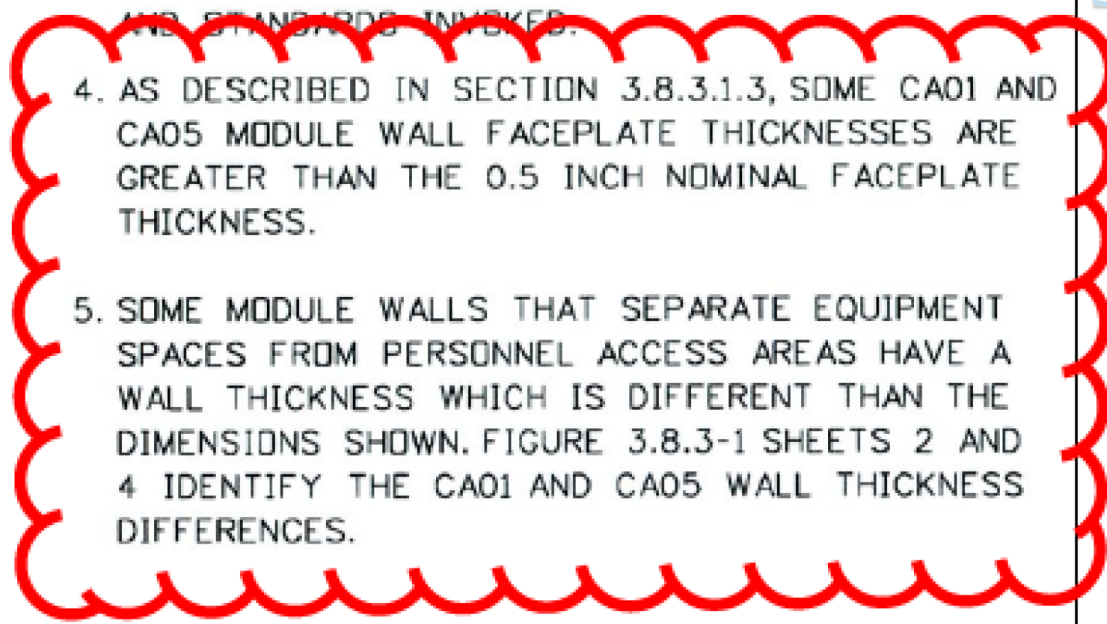
[This figure contains Security Related Information. See Enclosure 4]

UFSAR Figure 3.8.3-1, (Sheet 5 of 7) Structural Modules in Containment Internal Structures – Revise to add a note stating “Note: View shows module elevation 135’-3”” as shown in Figure 5 below:

Figure 5

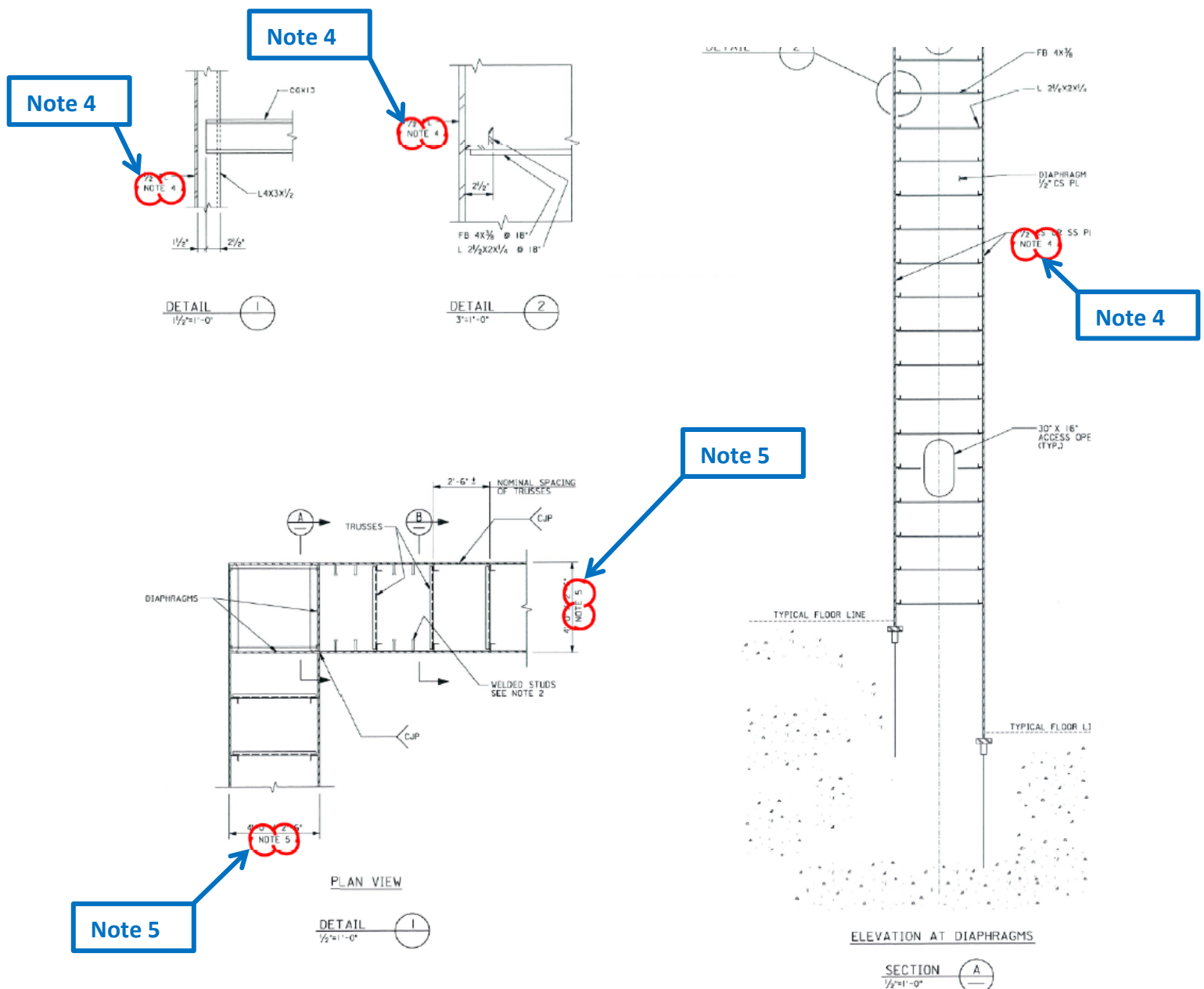


UFSAR Figure 3.8.3-8, Structural Modules – Typical Design Details, (Sheet 1 of 3) – Revise to add the following Notes 4 and 5 as shown below:



UFSAR Figure 3.8.3-8, Structural Modules – Typical Design Details, (Sheet 1 of 3) – Revise to add the appropriate call-outs as shown in Figure 6:

Figure 6



UFSAR Figure 3.8.3-8, Structural Modules – Typical Design Details, (Sheet 2 of 3) – Revise to add the two notes as shown in the Figure 7 below:

Figure 7

NOTES:

THE TYPICAL DETAILS SHOWN REPRESENT THE FUNDAMENTAL APPROACH FOR THE COMPOSITE WALL MODULES. THE FINAL DESIGN DETAILS MAY DIFFER FROM THOSE SHOWN FOR THE FOLLOWING REASONS.

- ACCESSIBILITY FOR INSPECTION DURING FABRICATION AND CONSTRUCTION
- LESSONS LEARNED FROM IMPLEMENTATION OF THE DESIGN
- EASE OF FABRICATION AND CONSTRUCTION
- RESOLUTION OF CONSTRUCTABILITY ISSUES AND SEQUENCES
- VARIATION IN MODULE SIZE AND CONFIGURATION CHANGES MADE DURING DETAILED DESIGN ARE IN ACCORDANCE WITH THE SPECIFIC CODES AND STANDARDS INVOKED.
- AS DESCRIBED IN SECTION 3.8.3.1.3, SOME CA01 AND CA05 MODULE WALL FACEPLATE THICKNESSES ARE GREATER THAN THE 0.5 INCH NOMINAL FACEPLATE THICKNESS.
- SOME MODULE WALLS THAT SEPARATE EQUIPMENT SPACES FROM PERSONNEL ACCESS AREAS HAVE A WALL THICKNESS WHICH IS DIFFERENT THAN THE DIMENSIONS SHOWN. FIGURE 3.8.3-1 SHEETS 2 AND 4 IDENTIFY THE CA01 AND CA05 WALL THICKNESS DIFFERENCES.

UFSAR Figure 3.8.3-18, Location of Structural Wall Modules – Revise as shown in Figure 8 below:

Figure 8

